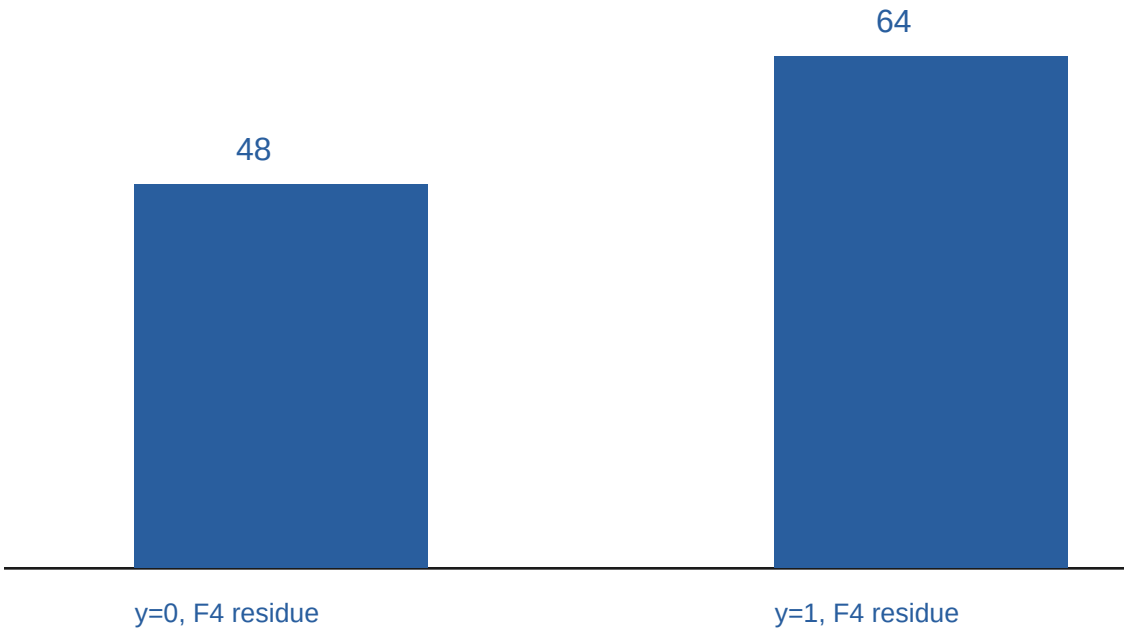
R101: marked-23 source-order fiber at x=1

5 November 2026 | Koblitz modular IC research

The SHA-256-bound R51 source gives f51(1,y)=y^3(y+1)^4 and G51(1,y,l)=l^16+l^8+1=(l^2+l+1)^8. All 28 rational lambda coefficients were accounted for. The lambda derivative vanishes at x=1. Singular checked both y=0 and y=1 lambda factors, full products, and changed-product controls. A fresh process rebound the donor and reproduced every export byte-for-byte. Independent GF(2) arithmetic and Lean 4.34.1 checked the polynomial and length identities.

F2 dimension of the x=1 source-order components



For F2[x,y,l]/(f51,G51), the x=1 fiber has two maximal ideals with F4 residue field. Local F2 dimensions are 48 and 64, totaling 112. These are source-order lengths; N51 was not used in this order’s fiber. Local normalization is required for branch counts and valuations. Attempts 01 and 02 were retained as preparation failures; attempt 03 and its fresh replay passed. The independent Python 3.9 verifier passed after its recorded bit-count compatibility correction.

Raw archive SHA-256: 6d672fe4f5ea9dc4ff66d7334586959cb120bb09e7521332bfcc85e2cc731c99